

Azure AI Foundry Agents (Solution Architect Track)

Azure AI Foundry supports agents, agent services, tool integration, evaluations, monitoring, and enterprise AI governance

Part 1 – What is Azure AI Foundry Agent Service?

Azure AI Foundry provides:

Models
Agents
Prompt Flow
Evaluations
Monitoring
Safety

These components are part of the Azure AI Foundry ecosystem.

Traditional Chatbot:

User
↓
GPT
↓
Response

Azure AI Foundry Agent:

User



Agent



Reasoning



Tools



Knowledge



Execution



Response

Part 2 – When Should You Use Agents?

Use GPT Only:

Simple Q&A

Content Generation

Summarization

Use RAG:

Knowledge Search

Policy Assistant

Document Search

Use Agents:

- Multi-Step Tasks
- Workflow Automation
- Task Execution
- Tool Calling
- Enterprise Operations

Example:

User:

Create exit request for employee.

Agent Actions:

- Retrieve employee details
- Generate checklist
- Create service desk ticket
- Generate email
- Notify manager

This aligns closely with Employee Exit Management scenarios you have worked on.

Part 3 – Agent Architecture

User



Teams / Web App



Azure AI Foundry Agent



Model



Tools



Knowledge Sources



Business Systems

Part 4 – Core Components

1. Agent

Defines:

- Behavior
- Purpose
- Instructions

Example:

You are an HR assistant.

Always answer policy-related questions.

Provide official responses only.

2. Model

Examples:

GPT-4o

GPT-4.1

GPT-4o-mini

Used as the reasoning engine.

3. Instructions

System Prompt

Example:

You are an Employee Exit Assistant.

Be concise.

Use company policy documents.

Instructions are a core part of agent definition and configuration.

4. Thread

A thread maintains conversation context.

User:

What is my leave balance?

User:

Apply leave tomorrow.

User:

Notify manager.

The thread maintains continuity.

Part 5 – Agent Lifecycle

Create Agent



Configure Instructions



Attach Tools



Attach Knowledge



Create Thread



User Interaction



Response

Part 6 – Built-In Tools

Common Agent Tools:

File Search

Search Documents

Search PDFs

Search Knowledge Base

Code Interpreter

Data Analysis

Report Creation

Calculations

Knowledge Retrieval

Azure AI Search

Enterprise Search

Agent services support tool extension and enterprise integrations.

Part 7 – Custom Tools

Custom tools can connect to:

Azure Functions

REST APIs

SQL Database

GitHub

Azure DevOps

ServiceNow

SharePoint

Example:

Check Leave Balance()

Create Ticket()

Get Employee Details()

Part 8 – Knowledge Sources

Without knowledge:

GPT only knows training data.

With Knowledge:

SharePoint
Azure AI Search
Runbooks
Policies
Project Documents

Knowledge-enhanced agents are a key Azure AI Foundry capability.

Part 9 – Architect View

As a Solution Architect, don't think:

How do I build an Agent?

Think:

How do I design enterprise agent ecosystems?

Questions:

- What tools are required?
 - What systems are accessed?
 - What security controls exist?
 - What monitoring is required?
 - What costs are involved?
 - How is governance enforced?
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Part 10 – Security Architecture

Authentication

Managed Identity

Preferred for enterprise deployments.

Authorization

Azure RBAC

Secrets

Azure Key Vault

Networking

Private Endpoints

Part 11 – Enterprise Reference Architecture

Employee



Teams



Azure AI Foundry Agent



Azure OpenAI



Azure AI Search



SharePoint



Enterprise Systems

└─ HRMS

└─ ServiceNow

└─ Azure DevOps

└─ SQL Database



Response

Hands-On Assignment

Scenario

Design an Employee Exit Assistant using Azure AI Foundry.

Requirements:

Knowledge

Exit Policy

Asset Return Policy

Handover Guidelines

Tools

Create Ticket

Generate Email

Send Notification

Systems

HRMS

Service Desk

SharePoint

Security

Managed Identity

RBAC

Key Vault

Architecture Exercise

Draw this architecture:

Teams



AI Foundry Agent



Azure OpenAI



Azure AI Search



SharePoint Documents



HR System



Ticketing System

Add:

RBAC

Monitoring

Key Vault

Managed Identity

Interview Questions

1. What is Azure AI Foundry Agent Service?
2. What is the difference between Agent and RAG?
3. What is an Agent Thread?
4. What are Agent Tools?
5. What are Custom Tools?
6. How does an Agent access knowledge?
7. How would you secure an AI Agent?
8. Design an Employee Exit Agent.
9. Design an IT Helpdesk Agent.
10. Explain Agent Lifecycle.